

5. Diagrams **A**, **B**, **C** and **D** represent argon (Ar), nitrogen (N_2), oxygen (O_2) and carbon dioxide (CO_2), but not in that order.

**A****B****C****D**

- (a) Give the **letter** of the diagram that represents argon. [1]

.....

- (b) Give the **letter** of the diagram that represents a compound. Give a reason for your answer. [2]

Letter

Reason

- (c) (i) Use information from the diagrams above. Draw a diagram to represent a molecule of nitrogen dioxide, NO_2 . [1]

- (ii) Calculate the relative formula mass (M_r) of nitrogen dioxide, NO_2 . [1]

$$A_r(\text{N}) = 14$$

$$A_r(\text{O}) = 16$$

$$M_r = \dots\dots\dots$$

- (iii) Balance the equation for the reaction between nitrogen and oxygen to produce nitrogen dioxide. [1]

